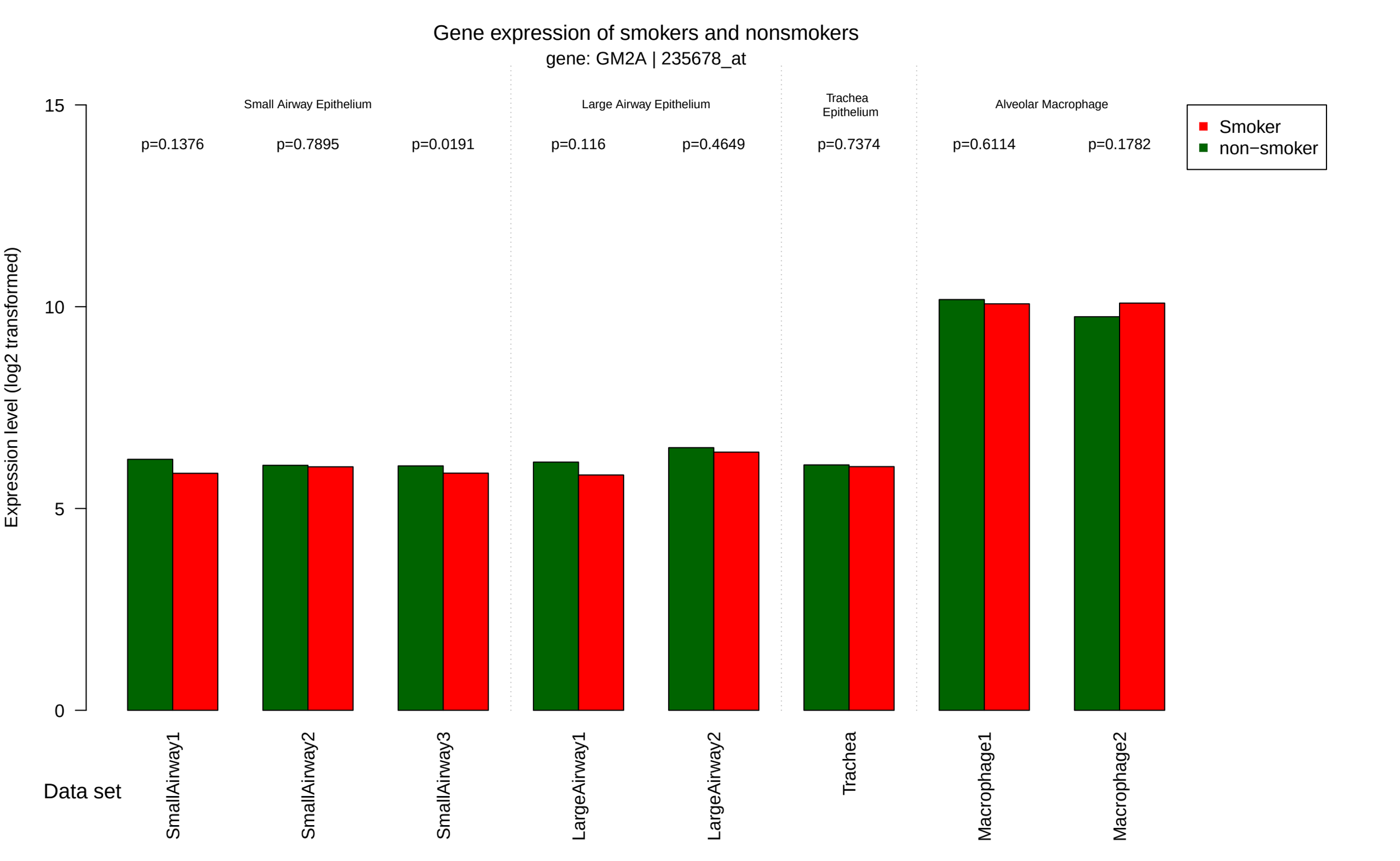
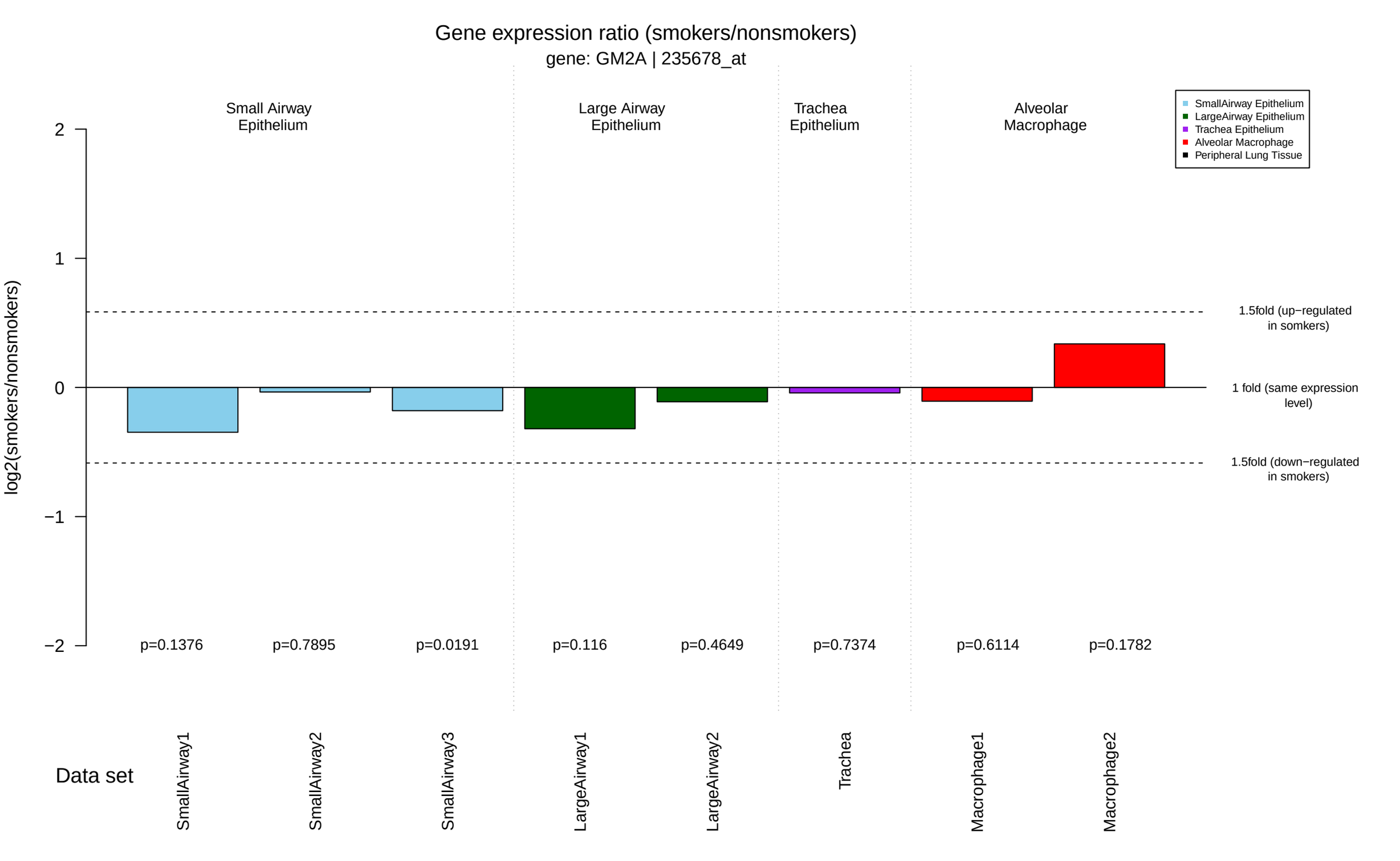
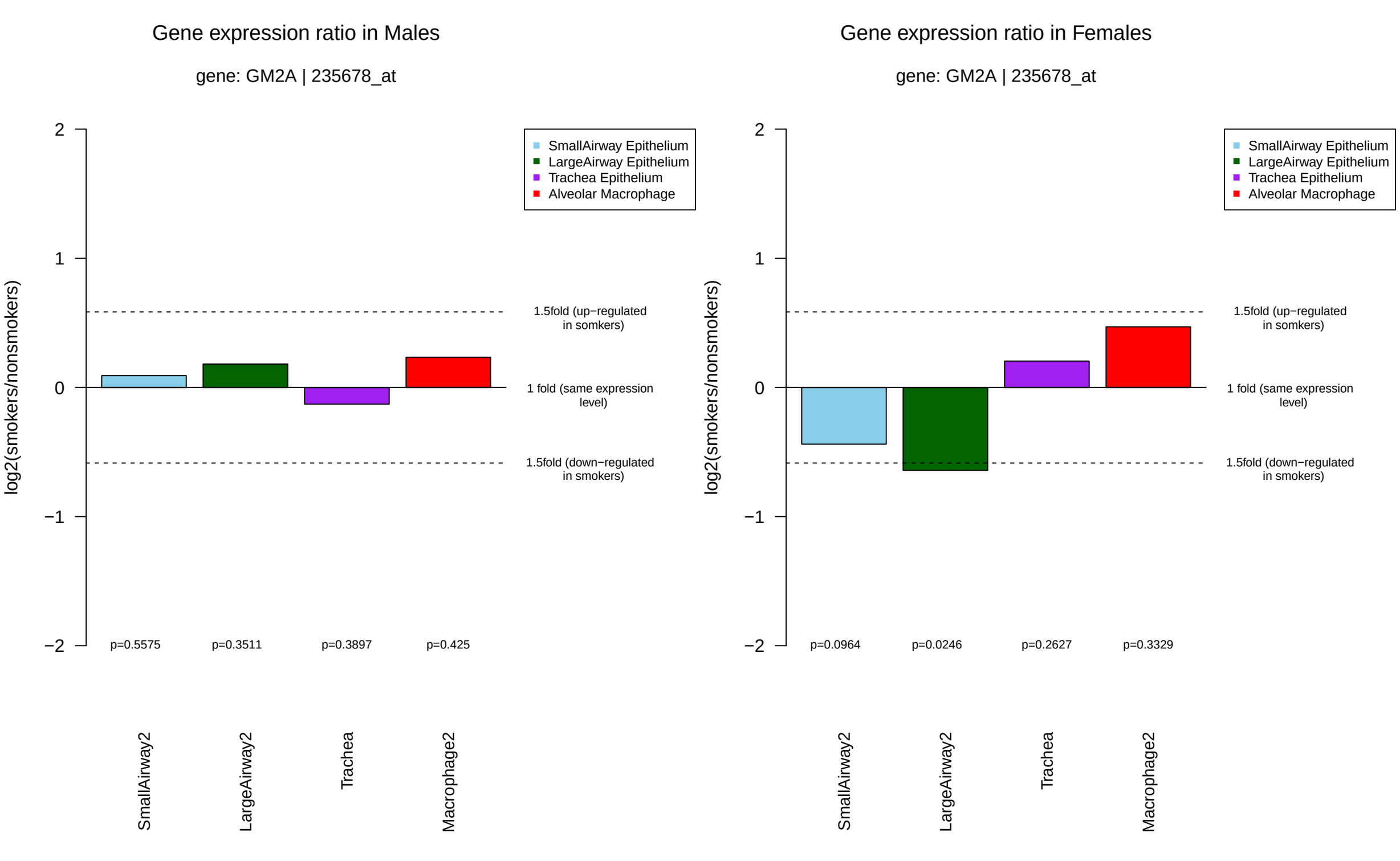
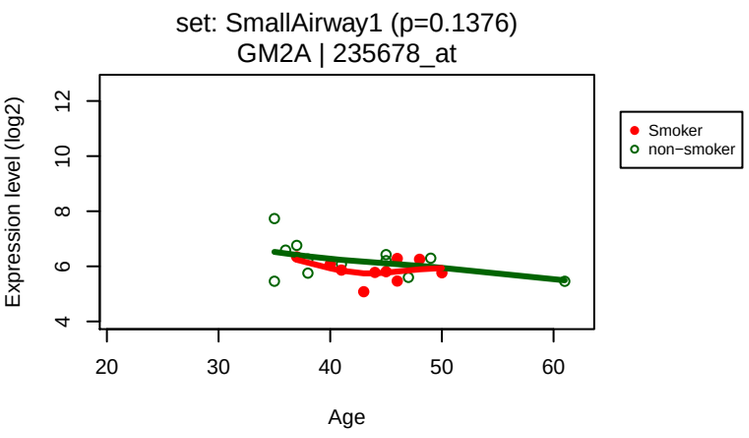
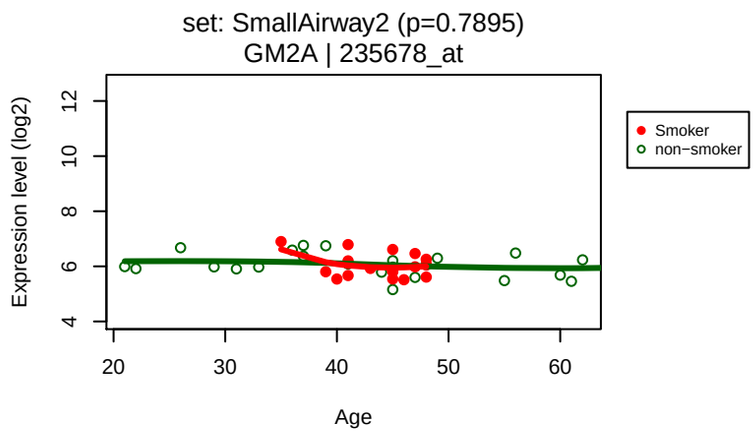
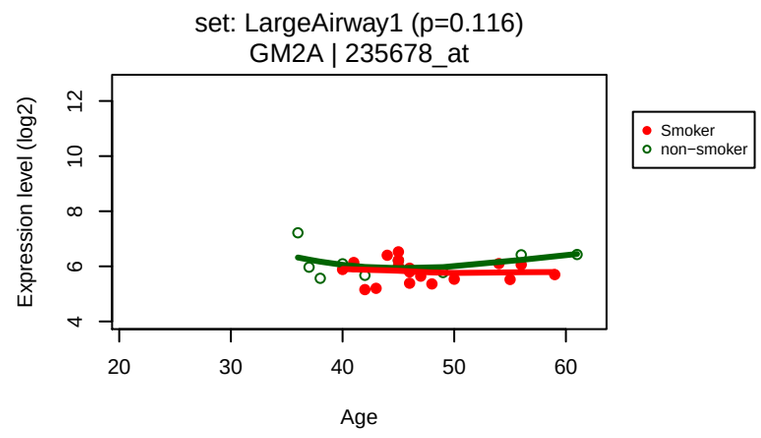

Gene expression of smokers and nonsmokers
gene: GM2A | 235678_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1376
p=0.7895
p=0.0191
p=0.116
p=0.4649
p=0.7374
p=0.6114
p=0.1782
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

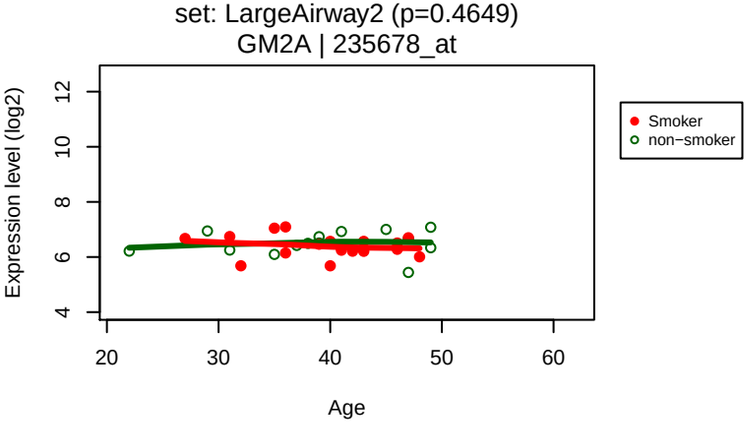

Gene expression ratio (smokers/nonsmokers)
gene: GM2A | 235678_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1376
p=0.7895
p=0.0191
p=0.116
p=0.4649
p=0.7374
p=0.6114
p=0.1782
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

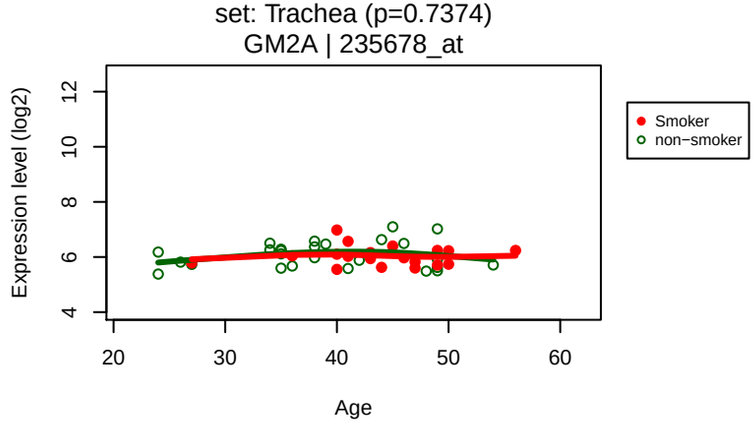

Gene expression ratio in Males
gene: GM2A | 235678_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5575
p=0.3511
p=0.3897
p=0.425
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GM2A | 235678_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0964
p=0.0246
p=0.2627
p=0.3329
SmallAirway2
LargeAirway2
Trachea
Macrophage2

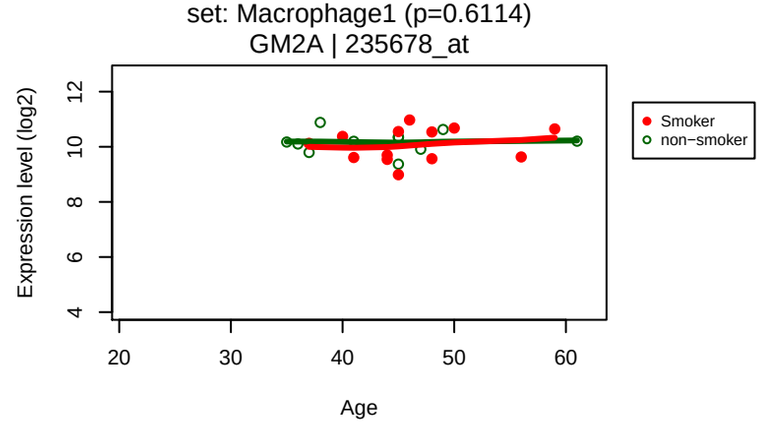

set: SmallAirway1 (p=0.1376)
GM2A | 235678_at
Expression level (log2)
Age
Smoker
non−smoker

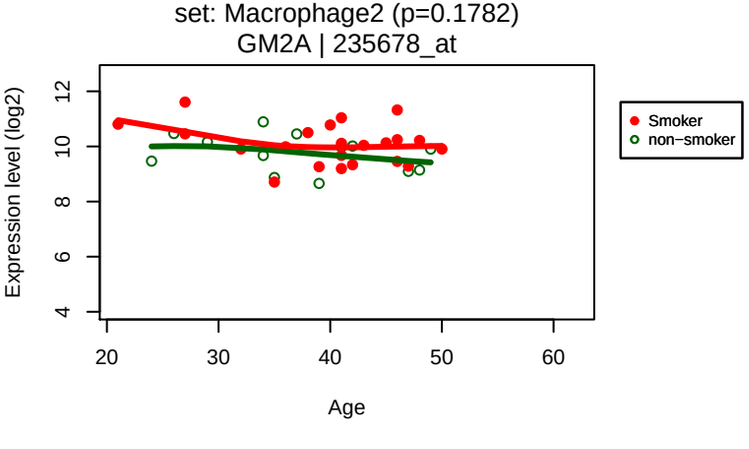

set: SmallAirway2 (p=0.7895)
GM2A | 235678_at
Expression level (log2)
Age
Smoker
non−smoker

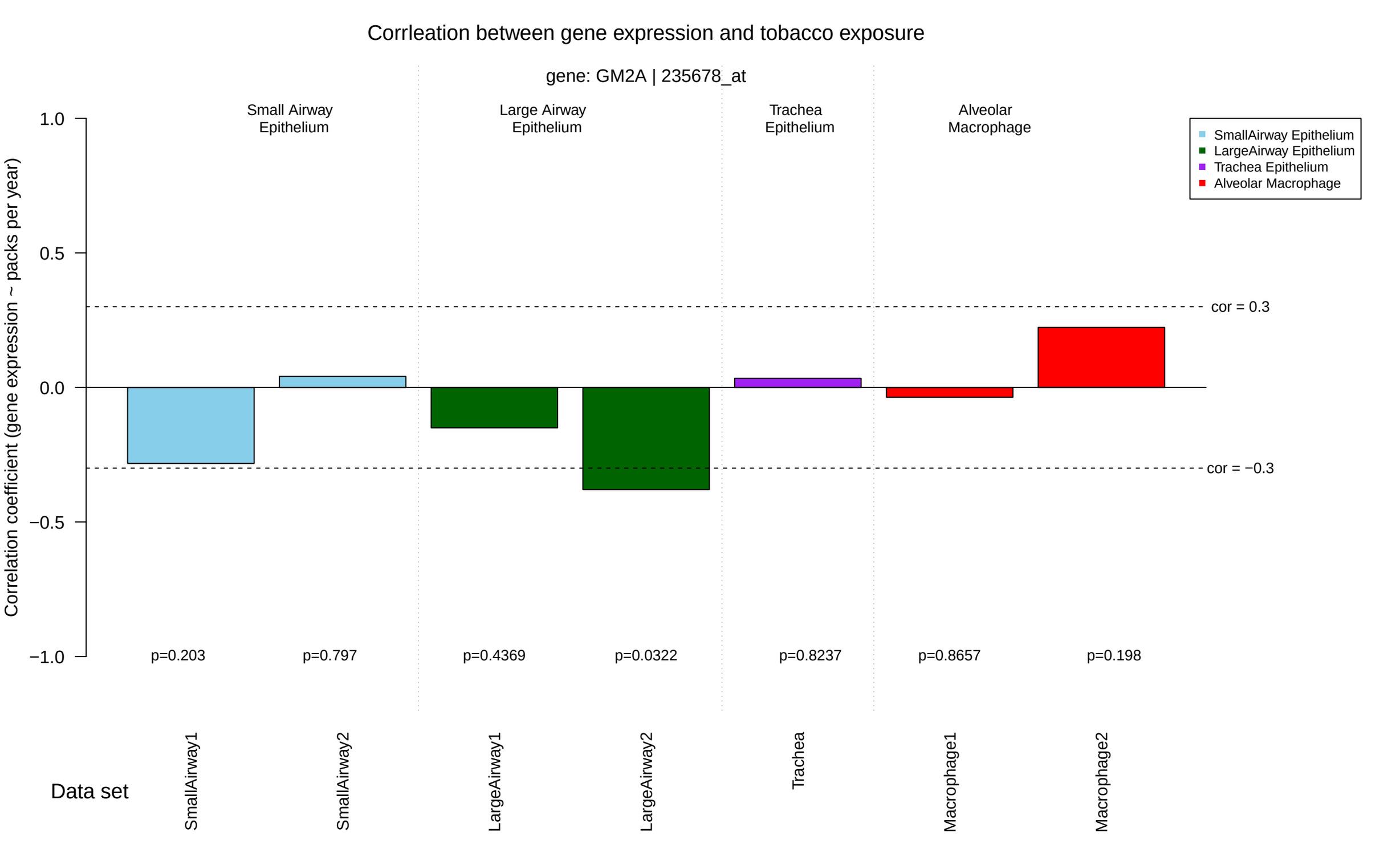

set: LargeAirway1 (p=0.116)
GM2A | 235678_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4649)
GM2A | 235678_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7374)
GM2A | 235678_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6114)
GM2A | 235678_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1782)
GM2A | 235678_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GM2A | 235678_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.203
p=0.797
p=0.4369
p=0.0322
p=0.8237
p=0.8657
p=0.198
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2